

Notes: Chen, Chen & Yang (RES, 2026)
Women in the Courtroom: Technology and Justice

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1 Big picture

- **Research question.** Can information technology and public monitoring reduce gender disparities in judicial decisions?
- **Setting.** China implemented an open justice reform that required courts to broadcast courtroom proceedings live and archive recordings on China Court Trial Online.
- **Data scale.** The paper studies roughly 6.4 million civil judgments involving individual litigants from 2014 to 2018, linked to broadcast records and region-level covariates.
- **Baseline disparity.** Before the reform, female plaintiffs were less likely to win than male plaintiffs, conditional on rich case controls, judge fixed effects, court-area fixed effects, and time fixed effects.
- **Main empirical design.** The authors use a generalized difference-in-differences design with continuous treatment: broadcast intensity at the court-area by year-quarter level.
- **Main result.** Higher broadcast intensity reduces the female plaintiff disadvantage in winning probability.
- **Endogeneity concern.** Cases selected for broadcasting are not random. The paper therefore avoids case-level treatment and uses aggregate intensity; it also implements a Bartik-style IV based on pre-reform case-area shares and leave-one-out area-by-prefecture broadcast intensity.
- **Mechanism.** Textual analysis of judgments suggests that transparency changes judge behavior: judges acknowledge female plaintiffs' claims more, cite more legal articles, and write less standardized, more detailed opinions.
- **Alternative channels.** Results hold in samples where litigants or lawyers do not appear in court, suggesting the primary mechanism is judge behavior rather than strategic behavior by litigants or lawyers.
- **Contribution.** The paper links gender bias, judicial transparency, public-sector monitoring, and information technology using unusually large-scale court text data.
- **Policy tension.** Cameras and public broadcasts can improve accountability, but they may also create privacy costs for litigants.

2 Institutional background and data construction

2.1 China's civil court system and why judicial discretion matters

- China's courts are hierarchical: local courts, intermediate courts, high courts, and the Supreme People's Court.
- Civil cases are divided into nine major areas, including personal rights, marriage/family/inheritance, property, contracts, intellectual property, industrial disputes, finance/security/insurance, tort liability, and special procedures.
- Many simple civil cases are heard by one judge under summary procedure. More complex cases can involve a panel of three, five, or seven judges, but the chief judge remains influential.
- Judges therefore have meaningful discretion in how they assess litigant claims, recognize facts, apply legal articles, and decide outcomes.

Interpretation: The judicial setting makes the reform relevant: if a single judge's effort, attention, or bias can affect outcomes, increasing observability of judge behavior can plausibly change decisions.

2.2 Open justice reform and live trial broadcasting

- The Supreme People’s Court launched a set of transparency initiatives beginning in 2014.
- The key reform studied here is the introduction of courtroom cameras and live trial broadcasts on China Court Trial Online.
- The platform officially launched in September 2016 and allowed the public to watch live proceedings or view archived recordings.
- Implementation varied by timing and intensity across courts. Some courts connected early, while others joined later; courts also differed in what fraction of cases they broadcast.
- Lower-level courts did not necessarily have the same capacity or enthusiasm as the Supreme Court. Higher courts used quotas, performance evaluation, honors, and rewards to induce broadcasting.

Interpretation: The reform creates quasi-experimental variation because courts were pushed to adopt the technology at different times and with different intensities. The variation is not purely random, but it is not designed to target female plaintiffs or gender disparities.

2.3 Judgment data from China Judgments Online

- The primary data source is China Judgments Online, the public repository of court rulings.
- The authors obtained legal documents with assistance from a commercial data company.
- The sample covers January 2014 to December 2018, which spans both the pre-reform period and the reform implementation period.
- The paper restricts attention to civil litigation involving individual litigants rather than institutions or companies.
- The full sample contains 6,424,324 civil judgments.
- The paper analyzes both: (i) the main sample containing all individual-litigant cases, including cases with multiple litigants on either side, and (ii) a two-litigant subsample with exactly one plaintiff and one defendant.

Interpretation: The main sample maximizes scale and precision. The two-litigant subsample provides cleaner gender measurement because the plaintiff and defendant identities are unambiguous.

2.4 Judgment structure and variables extracted from text

A typical judgment contains five sections:

1. basic case information;
2. claims of litigants;
3. facts recognized by the court;
4. legal principles applied;
5. outcome of the litigation.

The authors extract:

- court, case area, dates, and case number;
- plaintiff and defendant characteristics;
- judge names and chief judge information;

- whether parties appeared in court;
- numbers of plaintiffs, defendants, plaintiff lawyers, and defendant lawyers;
- legal costs, interpreted as a proxy for case size;
- case type and litigation instance;
- the outcome, especially the plaintiff’s chances of winning.

Interpretation: The paper uses the textual richness of Chinese court decisions not only to identify outcomes, but also to infer judicial attention and effort.

2.5 Gender measurement

- Plaintiff and defendant gender are inferred from litigant names and other document information.
- Judge gender is inferred from names using the Ngender algorithm and validated against a database of nearly 2,000 judges from Supreme Court merit lists.
- The validation exercise gives about 92% accuracy for judge gender prediction.

Interpretation: Gender inference is not perfect, but validation suggests the measurement error is not overwhelming. The two-litigant subsample further reduces ambiguity in plaintiff and defendant gender.

2.6 Outcome variable: plaintiff chances of winning

- The main outcome is the extent to which the plaintiff’s claims are supported by the court.
- It is based on the outcome section of each judgment, including allocation of legal costs between plaintiff and defendant.
- The average plaintiff winning rate is about 74% in the main sample.

Interpretation: The outcome is not simply a binary win/loss measure. It is closer to a continuous measure of judicial support for the plaintiff’s claims, which is well suited for civil cases where outcomes can be partial.

2.7 Treatment variable: broadcast intensity

- The authors do not use a case-level indicator for whether a case is broadcast, because case selection into broadcasting is endogenous.
- Instead, they construct broadcast intensity at the court-area by year-quarter level:

$$Intensity_{jkt} = \frac{\text{number of broadcast cases in area } j, \text{ court } k, \text{ quarter } t}{\text{total cases in area } j, \text{ court } k, \text{ quarter } t}.$$

- The default aggregation level is court-area by year-quarter.
- Figure 1 shows that most courts broadcast only a small share of cases, but intensity varies substantially.
- Area-level intensity also varies: marriage/family/inheritance disputes tend to be lower intensity, while intellectual property disputes tend to be higher intensity.

Interpretation: Aggregate intensity is intended to capture how much the courtroom environment becomes visible and monitored. It reduces the direct selection problem that arises when courts choose exactly which individual cases to broadcast.

2.8 Controls and supplementary regional data

Case-level controls include:

- defendant gender;
- numbers of judges, plaintiffs, defendants, plaintiff lawyers, and defendant lawyers;
- whether plaintiffs and defendants appeared in court;
- litigation instance;
- log legal costs;
- case type.

Prefecture-level controls include:

- internet penetration rate;
- GDP per capita;
- population.

Interpretation: These controls absorb observed case complexity, litigant composition, local development, and internet access, all of which could be related to both broadcasting intensity and litigation outcomes.

2.9 Mechanism variables from judgment text

- **Text similarity:** the authors compare the plaintiff’s statement, the defendant’s statement, and facts recognized by the judge. Higher similarity to a litigant’s statement proxies judicial attention to that litigant.
- **Δ similarity:** plaintiff-similarity minus defendant-similarity. This captures relative attention to plaintiff claims.
- **Number of legal articles cited:** more citations indicate more effort in legal reasoning.
- **Writing-style entropy:** a higher entropy index indicates less standardized, more case-specific writing.

Interpretation: The mechanism tests are important because the main outcome result alone cannot distinguish whether transparency changes judges, litigants, lawyers, or case selection. Text measures make judge behavior observable.

3 Empirical specifications and interpretation

3.1 Pre-reform gender disparity specification

For the pre-reform period 2014Q1 to 2016Q2, the paper estimates:

$$y_{ijkl} = \beta_0 + \beta Female_i + X_i' \zeta + \omega_{j \times k} + \lambda_t + \delta_l + \varepsilon_{ijkl}.$$

- y_{ijkl} is the plaintiff’s chance of winning in case i , area j , court k , quarter t , judge l .
- $Female_i$ equals one if the plaintiff is female.
- X_i contains case and prefecture controls.
- $\omega_{j \times k}$ are court-area fixed effects.
- λ_t are year-quarter fixed effects.
- δ_l are judge fixed effects.

Interpretation: The coefficient β measures the female plaintiff disadvantage within the same court-area, same time period, and same judge, after controlling for observable case differences. It is not interpreted as causal gender discrimination by itself, because unobserved case differences may remain.

3.2 Continuous difference-in-differences specification

For the full 2014-2018 period, the main specification is:

$$y_{ijkl} = \beta_0 + \beta Female_i + \gamma Intensity_{jkt} + \theta(Female_i \times Intensity_{jkt}) + X_i' \zeta + \omega_{j \times k} + \lambda_t + \delta_l + \varepsilon_{ijkl}.$$

- The treatment is continuous broadcast intensity, not a simple post dummy.
- The key coefficient is θ on $Female_i \times Intensity_{jkt}$.
- A positive θ means female plaintiffs do better relative to male plaintiffs as transparency intensity rises.

Interpretation: The DID compares the gender gap across court-area cells and over time as broadcast intensity changes. Because it focuses on the interaction with female plaintiff status, it asks whether transparency narrows the gender gap rather than whether transparency raises all plaintiff win rates.

3.3 Event-study specification

The paper estimates dynamic effects around prefecture-level reform timing:

$$y_{ijkl} = \beta_0 + \beta Female_i + \sum_{\tau=-6}^8 \gamma_{\tau} Reform_{p,t}^{\tau} + \sum_{\tau=-6}^8 \theta_{\tau}(Female_i \times Reform_{p,t}^{\tau}) + X_i' \zeta + \omega_{j \times k} + \lambda_t + \delta_l + \varepsilon_{ijkl}.$$

- $Reform_{p,t}^{\tau}$ indicates that quarter t is τ quarters away from the reform introduction in prefecture p .
- Quarter $\tau = -1$ is the omitted reference period.
- The coefficients θ_{τ} trace the evolution of the female-male gap over time.

Interpretation: The event study checks whether there are pre-trends and whether effects grow gradually after the reform. Gradual strengthening is expected because broadcast intensity rises over time rather than jumping immediately to a high level.

3.4 Bartik instrumental-variable specification

The Bartik instrument is constructed as:

$$Z_{kt} = \sum_{j \in Areas} \alpha_{jk}^{2014} \overline{Intensity}_{jp(-k),t},$$

where α_{jk}^{2014} is court k 's 2014 pre-reform share of cases in area j , and $\overline{Intensity}_{jp(-k),t}$ is the average intensity for area j in prefecture p at time t , excluding the focal court.

- Z_{kt} predicts how strongly a court is exposed to the reform because of its pre-existing area composition.
- The instrument uses variation in other courts' area-specific broadcasting intensity within the same prefecture.
- First stages use Z_{kt} and $Female_i \times Z_{kt}$ to instrument $Intensity_{jkt}$ and $Female_i \times Intensity_{jkt}$.

Interpretation: The IV strategy addresses the concern that realized intensity may be correlated with unobserved court-area shocks. It relies on pre-reform case-area shares being conditionally exogenous to post-reform gender-specific outcome trends.

3.5 Mechanism specifications

The mechanism tests replace the dependent variable with:

- $\Delta\textit{Similarity}$, measuring relative judicial attention to plaintiff claims;
- number of legal articles cited;
- writing-style entropy;
- plaintiff winning chances interacted with entropy.

Interpretation: These specifications ask whether transparency changes the process of judging, not only the final outcome. A core claim of the paper is that the reform makes judges more attentive and effortful, especially in cases involving female plaintiffs.

3.6 Judge-gender and alternative-channel specifications

- The judge-gender analysis estimates the main DID separately for female and male chief judges.
- The alternative-channel analysis restricts the sample to cases where litigants or lawyers do not appear in court.

Interpretation: If the result comes primarily from judge behavior, it should remain when litigants or lawyers are absent. If the result comes from litigants or lawyers changing courtroom behavior, the effect should weaken sharply in those samples.

4 Identification strategy in detail

4.1 Step 1: Document a pre-reform gender gap, but do not over-interpret it causally

The pre-reform regression shows female plaintiffs are less likely to win. The authors include rich controls and judge fixed effects, but unobserved case content can still differ by plaintiff gender. This is why the pre-reform estimate is best interpreted as evidence of a gender disparity, not as definitive causal proof of bias.

4.2 Step 2: Use the open justice reform as a monitoring shock

The reform introduced public observability through trial broadcasting. The key identifying intuition is that a monitoring shock should reduce discretionary or biased behavior by judges, while a purely case-composition story should not necessarily create gender-gap reductions tied to broadcast intensity.

4.3 Step 3: Avoid case-level broadcast selection

Courts select which cases to broadcast, and Table A2 shows that observable case characteristics predict broadcasting. Therefore, a naive comparison of broadcast versus non-broadcast cases would be biased. The paper instead uses aggregate broadcast intensity at the court-area-quarter level.

4.4 Step 4: Use court-area and time fixed effects

Court-area fixed effects absorb stable differences across courts and legal areas. Year-quarter fixed effects absorb national time shocks. Judge fixed effects control for stable judge-specific differences. The identifying variation comes from changes in broadcast intensity within court-area cells over time.

4.5 Step 5: Test dynamics

The event study checks whether gender-gap changes precede the reform. The paper reports positive and increasing post-reform coefficients, consistent with the gradual rollout and rising intensity of broadcasting.

4.6 Step 6: Implement Bartik IV

The Bartik IV uses pre-reform area shares and leave-one-out prefecture area intensity. This isolates predicted exposure based on pre-existing legal-area composition and broader local implementation patterns, rather than the focal court's own potentially endogenous choices.

4.7 Step 7: Probe the IV exclusion restriction

The paper discusses and implements checks motivated by the Bartik literature:

- placebo pre-trend tests using pre-reform outcomes and later predicted intensity;
- controls for interactions of time dummies with initial regional characteristics;
- balancing tests on prefecture controls;
- alternative forms of the instrument.

Interpretation: The IV is credible if 2014 area shares do not predict future gender-gap trends except through reform intensity. This is a stronger and more transparent identifying assumption than claiming realized broadcast intensity is random.

4.8 Step 8: Mechanism and falsification logic

Mechanism tests show that judges become more attentive and effortful. Alternative-channel tests show effects remain when litigants or lawyers are absent. Together, these results support the interpretation that the technology changes judge behavior.

4.9 Remaining threats

- Broadcast intensity may still be correlated with unobserved time-varying court quality.
- Gender inferred from names can have measurement error.
- Text similarity and entropy are proxies, not direct observation of cognition or effort.
- The reform may affect case filing behavior, though the paper reports evidence against large filing changes.
- The reform may have privacy and chilling costs not captured by litigation outcome data.

5 Tables and figures

5.1 Figure 1: Broadcast intensity across courts and areas

Read:

- The left panel shows the distribution of courts by post-reform broadcast intensity.
- More than 60% of courts broadcast fewer than 10% of their cases.
- Fewer than 20% of courts broadcast more than 20% of cases.

- The right panel shows variation by legal area: marriage/family/inheritance disputes have low broadcast intensity, while intellectual property disputes have higher intensity.

Interpretation: Figure 1 is important because it shows meaningful treatment variation. The reform is not simply a national post-period dummy; it varies by court and legal area. This supports the continuous-treatment DID design.

Economic message: The figure reveals that transparency was adopted unevenly, creating the empirical variation used to identify how monitoring intensity affects gender disparities.

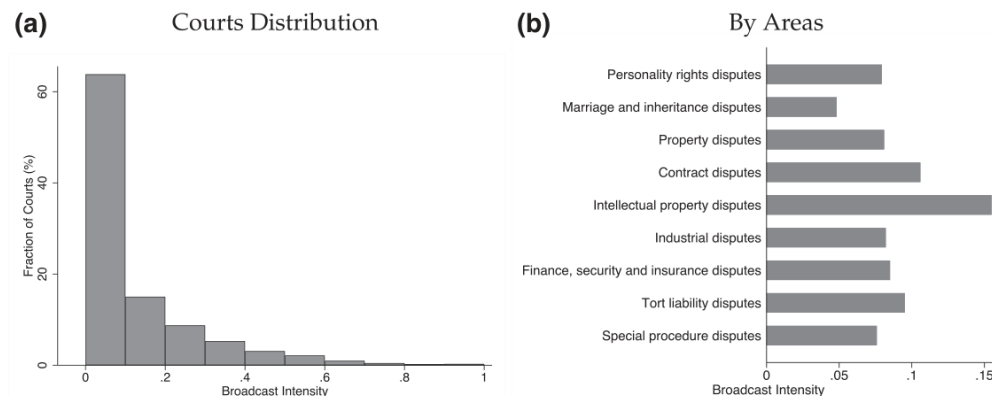


FIGURE 1

Broadcast intensity across courts and areas

Notes: The left panel displays the distribution of courts by broadcast intensity post-reform. The right panel shows the broadcast intensity across nine civil litigation areas. (a) Courts Distribution. (b) By Areas

5.2 Table 1: Plaintiff gender and litigation outcomes: the pre-reform period

Read:

- In the pre-reform period, female plaintiffs are less likely to win than male plaintiffs.
- The coefficient on Female is -0.0356 in the all-area main sample and -0.0530 in the all-area two-litigant subsample.
- Excluding marriage and family disputes reduces the magnitude but the coefficient remains negative and statistically significant.
- The mean plaintiff winning rate is 0.636 in the main all-area sample and 0.534 in the two-litigant all-area sample.

Interpretation: The table documents a gender disparity before the broadcasting reform. Because the specification includes judge fixed effects, the result compares male and female plaintiffs before the same judges. However, it still cannot prove discrimination because female and male plaintiffs may differ in unobserved case quality.

Economic message: The pre-reform disparity motivates the reform analysis. The key empirical question is whether increased observability reduces this gap.

TABLE 1
Plaintiff gender and litigation outcomes: the pre-reform period

	Outcome Variable: Chances of Winning			
	All Litigants		Sub-sample: Two Litigants	
	All Areas (1)	Excl Marriage (2)	All Areas (3)	Excl Marriage (4)
Female	-0.0356*** (0.00156)	-0.0226*** (0.00159)	-0.0530*** (0.00229)	-0.0344*** (0.00233)
Controls	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y
Mean of outcome	0.636	0.730	0.534	0.680
N	1,070,802	883,115	614,414	436,647
R ²	0.581	0.486	0.657	0.609

Notes: This table presents the results of analyzing gender differentials in the plaintiff's chances of winning for civil cases adjudicated during the pre-reform period from 2014Q1 to 2016Q2. Columns (1) and (2) report the results with main sample and columns (3) and (4) present the results with two-litigant subsample. In columns (2) and (4), we further exclude cases in the area of marriage and family disputes. Case-level control variables include the defendant gender, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

effects, we find that the chances of winning for female plaintiffs are approximately 3.56 percentage points lower than for male plaintiffs (equivalent to 5.6% of the sample mean). This gender difference is even more pronounced in the two-litigant scenarios, where it is approximately 5.3 percentage points (equivalent to 10% of the sample mean). This difference is expected, as our full-sample estimations only consider the first litigant's gender, and gender mixing on either side likely biases estimates toward zero.

To ensure that the association is not driven only by gender-salient issues, such as litigation related to marriage, family and inheritance, we drop cases in this area and repeat our estimations with both samples. The results are reported in columns (2) and (4) of Table 1. Both estimated coefficients are smaller in size but still significant, suggesting that such an association exists in areas that are not gender salient.

4.2. Gender disparities and reform implementation

TABLE 2
DID estimation: impacts of live broadcasting

	Outcome Variable: Chances of Winning			
	All Litigants		Sub-sample: Two Litigants	
	All Areas (1)	Excl Marriage (2)	All Areas (3)	Excl Marriage (4)
Female $\times$ Intensity	0.0394*** (0.00255)	0.0249*** (0.00247)	0.0536*** (0.00336)	0.0302*** (0.00306)
Female	-0.0116** (0.000847)	-0.00455*** (0.000822)	-0.0199*** (0.00108)	-0.00943 (0.00101)
Intensity	-0.0177*** (0.00441)	-0.0117*** (0.00450)	-0.0178*** (0.00727)	-0.00993 (0.00756)
Controls	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y
Mean of outcome	0.744	0.787	0.709	0.788
N	3,974,316	3,696,048	2,063,379	1,810,190
R ²	0.464	0.386	0.584	0.495

Notes: This table presents the main results for the DID specification, using the sample of cases adjudicated between 2014 and 2018 (both the pre- and post-reform periods). Columns (1) and (2) report the results with main sample and columns (3) and (4) report the results with the two-litigant subsample. In columns (2) and (4), we exclude cases in the area of marriage and family disputes. Case-level control variables include the defendant gender, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

gender gap as litigation becomes more transparent provides evidence of the existence of gender discrimination.¹⁰

Dynamics Effects Beyond examining the reform's average effects, we also investigate its dynamic impact over time. As broadcast intensity gradually increased after the reform's introduction, we expect to observe a progressive strengthening of the reform's impact, rather than an abrupt, discontinuous change.

To investigate, we employ an event study model and exploit the variation in the timing of the reform across prefectures (*i.e.* the extensive margin), the administrative level at which the reform was decided, financed and implemented. We construct a dummy variable $\text{Reform}_{p,t}$, which takes a value of 1 if prefecture p (where court k is located) started the reform in its jurisdiction in quarter t , and 0 otherwise. To estimate the dynamic effects, we employ the following specification:

5.3 Table 2: DID estimation: impacts of live broadcasting

Read:

- The key coefficient is Female $\times$ Intensity.
- The coefficient is positive in all four columns: 0.0394, 0.0249, 0.0536, and 0.0302.
- The estimates are highly significant and remain positive when excluding marriage and family disputes.
- The two-litigant estimates are larger, suggesting the effect is clearer when plaintiff and defendant gender are measured cleanly.

Interpretation: Higher broadcast intensity narrows the gap between female and male plaintiffs. Because the baseline Female coefficient remains negative, the reform does not necessarily eliminate the gap, but it meaningfully reduces it.

Economic message: Public monitoring changes the distribution of judicial support across litigants, not just average outcomes. This is evidence that transparency can discipline gender-biased decision-making.

5.4 Figure 2: Event Study

Read:

- The event study plots the dynamic effect of the reform on the gender differential from six quarters before to eight quarters after reform introduction.
- The omitted period is the quarter immediately before the reform.
- Post-reform coefficients are positive and tend to increase over time.
- The gradual pattern matches the fact that broadcast intensity itself increased gradually after adoption.

Interpretation: The event study is a timing check. The absence of large negative pre-period movements and the strengthening of post-period estimates support the interpretation that the reform, rather than unrelated pre-existing trends, drives the observed narrowing of the gender gap.

Economic message: Transparency takes time to become effective because implementation and judge adaptation are gradual.

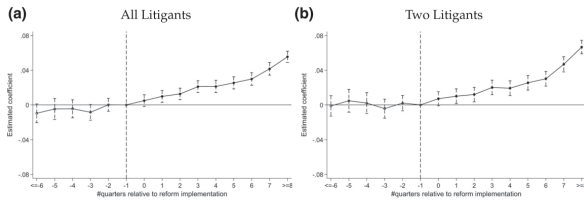


FIGURE 2
Event Study
Notes: The figure illustrates the reform's estimated effects on the gender differential in winning chances over time, spanning from six quarters before its introduction ($t = -6, \dots, -1, 0$) to eight quarters and beyond afterwards ($t = 1, \dots, 8$). Dots represent the estimated coefficients for each period, with bars indicating the corresponding 95 confidence intervals. (a) All Litigants. (b) Two Litigants

Outcome Variable	Bartik Instrument Estimation					
	All Litigants			Sub-sample: Two Litigants		
	1st Stage		2nd Stage	1st Stage		2nd Stage
	Intensity (1)	Female $\times$ Intensity (2)	Chances of Winning (3)	Intensity (4)	Female $\times$ Intensity (5)	Chances of Winning (6)
Female $\times$ Intensity			0.0726*** (0.00599)			0.0889*** (0.00699)
Female	0.000375 (0.000265)	0.0224*** (0.00163)	-0.0138*** (0.000903)	0.000503* (0.000293)	0.0217*** (0.00168)	-0.0224*** (0.00115)
Intensity			-0.0207*** (0.00677)			-0.0198*** (0.00724)
Predicted Intensity	0.687*** (0.0451)	-0.0799*** (0.00796)		0.703*** (0.0447)	-0.0745*** (0.00767)	
Female $\times$ Predicted Intensity	-0.0150*** (0.00550)	0.960*** (0.0360)		-0.0161*** (0.00556)	0.983*** (0.0347)	
Controls	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y
K-P LM Value			137.923			122.984
K-P Wald Value			114.858			122.643
N	3,974,316	3,974,316	3,974,316	2,063,379	2,063,379	2,063,379
R ²	0.739	0.565	0.017	0.765	0.592	0.026

Notes: This table presents the first stage and second stage results for the IV regressions. We construct a Bartik instrument by using the early shares of areas in each court in 2014 and the time-varying prefecture-level average broadcasting intensity for each area (excluding the focal court). For the main sample, columns (1) and (2) present results from the first stage and column (3) shows our estimate of interest from the second stage. For the two-litigant sample, the corresponding results from the first and second stages are shown in columns (4), (5), and (6). Case-level control variables include the defendant gender, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 reports our IV estimates using both samples. For the main sample, columns (1) and (2) of Table 3 present the first-stage results and column (3) displays our key estimate of interest from the second stage. For the two-litigant sample, the corresponding results from the first and second stages are shown in columns (4), (5), and (6). The coefficients in the first stage are highly significant, implying that the instrument has high predictive power. As shown in columns (1) and (4), each percentage increase in predicted broadcasting intensity is associated with a 0.688 and 0.703 two-litigant sample, respectively. The LM and Wald tests imply that underidentification and weak identification are unlikely to be concerns for our IV estimations.

The IV estimates for the interaction term shown in columns (3) and (6) remain positive and significant. We observe that the IV estimates are only slightly larger than the corresponding

5.5 Table 3: Instrumental variable estimations

Read:

- The first-stage coefficients show that predicted intensity is strongly related to actual intensity.
- The second-stage coefficient on Female $\times$ Intensity is 0.0726 in the main sample and 0.0889 in the two-litigant sample.
- These IV estimates are somewhat larger than the DID estimates but point in the same direction.
- The similarity between IV and DID estimates suggests that endogeneity in realized intensity is not overturning the baseline conclusion.

Interpretation: Table 3 strengthens identification. The Bartik instrument addresses the possibility that courts with certain unobserved trends choose higher broadcast intensity. The key identifying assumption is that 2014 area shares interact with other courts' subsequent area intensity only through the focal court's reform exposure.

Economic message: The IV confirms that increased monitoring reduces the female plaintiff disadvantage even when intensity is predicted by pre-reform area composition rather than the focal court's own choices.

5.6 Table 4: Text similarity, plaintiff gender, and the reform

Read:

- The dependent variable is Δ similarity: plaintiff-similarity minus defendant-similarity.
- In the pre-reform period, the Female coefficient is negative in both samples: judges acknowledge female plaintiffs' statements less than male plaintiffs' statements relative to defendants' statements.
- In the full-period DID, Female $\times$ Intensity is positive and significant.
- The two-litigant coefficient is especially large, 0.0159.

Interpretation: The table directly supports the attention mechanism. Before the reform, judges' recognized facts look less similar to female plaintiffs' statements. As broadcast intensity rises, this gap narrows.

Economic message: Monitoring appears to make judges take female plaintiffs' narratives more seriously, which can help explain why female plaintiffs' winning probability improves.

TABLE 4
Text similarity, plaintiff gender, and the reform

	Outcome Variable: Δ Similarity			
	Pre-reform Period		Full Period	
	All Litigants (1)	Two Litigants (2)	All Litigants (3)	Two Litigants (4)
Female	-0.0116*** (0.000718)	-0.0195*** (0.00106)	-0.00851*** (0.000491)	-0.0134*** (0.000703)
Intensity			-0.00177 (0.00658)	-0.00595 (0.00750)
Female $\times$ Intensity			0.00612*** (0.00226)	0.0159*** (0.00283)
Controls	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y
Mean of outcome	0.154	0.204	0.143	0.219
N	1,065,884	612,174	3,957,595	2,054,867
R ²	0.528	0.534	0.525	0.554

Notes: This table presents the regression results analyzing the relationship between text similarity, plaintiff gender and the reform. Δ similarity is measured as the difference between the similarity of the plaintiff's claim and the judge's claim, and the similarity of the defendant's claim and the judge's claim. The outcome variable measures the degree to which the judge acknowledges the plaintiff's statements compared to the defendant's statements. In column (1), we examine the correlation between Δ similarity and plaintiff gender, using the pre-reform subsample. In column (3), we further include broadcast intensity and the interaction term of female dummy and intensity. Consistent results are found in the two-litigant subsample, as presented in columns (2) and (4). Case-level control variables include the defendant gender dummy, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log) and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

First, we analyze the number of legal articles cited in judgments as an indicator of judicial

TABLE 5

Number of law articles cited and impacts of trials broadcasting

	Outcome Variable: Number of Law Articles Cited					
	All Litigants			Sub-sample: Two Litigants		
	Plaintiff's Gender:					
	All (1)	Female (2)	Male (3)	All (4)	Female (5)	Male (6)
Intensity	0.170*** (0.0629)	0.339*** (0.0873)	0.117* (0.0633)	0.170*** (0.0554)	0.275*** (0.0881)	0.127** (0.0572)
Controls	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y
Mean of outcome	8.663	9.490	8.289	6.536	7.356	6.145
N	3,974,316	1,210,518	2,745,275	2,063,379	637,567	1,404,989
R ²	0.678	0.706	0.677	0.524	0.577	0.529

Notes: This table presents results for the analysis of the number of law articles cited in the judgments. We examine the relationship between the number of law articles cited in the judgement and the reform intensity. The corresponding results are shown in column (1) for full sample, column (2) for female plaintiff sub-sample and column (3) for male plaintiff sub-sample. We also present the respective results with the two-litigant subsample in columns (4),(5), and (6). Case-level control variables include the defendant gender dummy, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log) and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

primarily driven by cases involving female plaintiffs. This finding indicates that increased judicial effort in cases with female plaintiffs may be a mechanism through which the reform reduces gender diversity in judicial decisions

5.7 Table 5: Number of law articles cited and impacts of trials broadcasting

Read:

- The dependent variable is the number of law articles cited in the judgment.
- Broadcast intensity increases the number of cited legal articles in the full sample and two-litigant sample.
- Effects are larger in female-plaintiff cases: 0.339 in the full sample and 0.275 in the two-litigant sample.
- Male-plaintiff coefficients are smaller, though still positive in some specifications.

Interpretation: Legal citations proxy judicial effort. The table suggests that transparency induces judges to provide more legal support in decisions, particularly in cases involving female plaintiffs.

Economic message: Under public scrutiny, judges appear to invest more in legal reasoning where the pre-reform disadvantage was strongest.

5.8 Table 6: Writing style, reform and litigation outcomes

Read:

- Columns (1)-(3) show that broadcast intensity raises writing-style entropy for all text, facts, and deliberation sections.
- Columns (4)-(6) show that Female $\times$ Entropy is positive and significant for winning chances.
- Higher entropy means less standardized, more case-specific writing.
- The positive interaction implies that more nuanced writing is associated with a smaller female plaintiff disadvantage.

Interpretation: The entropy results complement the law-article results. Judges are not only citing more law; their judgments become less formulaic and more detailed. The link between entropy and outcomes suggests that more careful deliberation is associated with more equal treatment.

Economic message: Transparency can improve the quality of decision-making by making judges write decisions that are more tailored to the case facts.

TABLE 6
Writing style, reform and litigation outcomes

	Outcome Variable:					
	Entropy for			Chances of Winning		
	All (1)	Facts (2)	Deliberation (3)	c = All (4)	c = Facts (5)	c = Deliberation (6)
Intensity	0.00495*** (0.00146)	0.00829* (0.00463)	0.00872* (0.00475)			
Female $\times$ Entropy _c				0.0663*** (0.0145)	0.0221*** (0.00760)	0.0497*** (0.00816)
Female				-0.0250*** (0.00364)	-0.0147*** (0.00223)	-0.0199*** (0.00195)
Entropy _c				-0.0329*** (0.0119)	0.00141 (0.00429)	0.00103 (0.00671)
Controls	Y	Y	Y	Y	Y	Y
Judge FE	N	N	N	Y	Y	Y
Court-Area FE	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y
Mean of outcome	0.314	0.358	0.363	0.744	0.744	0.744
N	258,995	258,995	258,995	3,965,940	3,965,940	3,965,940
R ²	0.422	0.212	0.359	0.462	0.462	0.462

Notes: This table presents results for the analysis of global entropy of judgements. We examine the correlation between entropy and broadcast intensity at the court-area-year-quarter level. We use three different entropy measures: Fact (factual descriptions), Deliberation (descriptions of deliberation), and ALL (combined measure of both parts). The corresponding results are displayed in columns (1), (2), and (3) for All, Facts, and Deliberation, respectively. In addition, we investigate whether the entropy level at the court-area-year-quarter level can predict gender disparity in litigation outcomes. To do so, we estimate equation (2) by substituting the intensity measure with the entropy measure. The results for the three types of entropy measures are presented in column (4) for All, column (5) for Fact, and column (6) for Deliberation, respectively. Case-level control variables include the defendant gender dummy, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. In all regressions, prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

for these aspects of the judgments: factual descriptions, deliberations, and a combined measure

	Outcome Variable: Chances of Winning			
	All Litigants		Sub-sample: Two Litigants	
	Chief Judge Gender:			
	Female (1)	Male (2)	Female (3)	Male (4)
Female $\times$ Intensity	0.0314*** (0.00398)	0.0424*** (0.00293)	0.0404*** (0.00492)	0.0588*** (0.00395)
Female	-0.00871*** (0.00106)	-0.0126*** (0.000921)	-0.0164*** (0.00139)	-0.0213*** (0.00120)
Intensity	-0.0199*** (0.00512)	-0.0167*** (0.00525)	-0.0237*** (0.00610)	-0.0154* (0.00932)
Controls	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y
Mean of outcome	0.747	0.743	0.721	0.705
N	1,144,191	2,826,878	583,122	1,476,902
R ²	0.454	0.474	0.572	0.595

Notes: This table shows the differential responses of female judges and male judges to the reform. We subdivide the sample into separate samples for female and male judges and re-estimate the baseline regression, as shown in columns (1) and (2) with the main sample and columns (3) and (4) with the two-litigant subsample. Case-level control variables include the defendant gender, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

where no litigants were present in court. This subsample is particularly useful because changes in litigant behaviour in response to the broadcasting mechanism would not be a factor in judicial decisions.¹⁷

We construct this subsample from our main sample and use it to estimate equation (2).

5.9 Table 7: Responses across judge genders

Read:

- The positive Female $\times$ Intensity coefficient appears for both female and male chief judges.
- The coefficient is larger for male judges: 0.0424 versus 0.0314 in the main sample, and 0.0588 versus 0.0404 in the two-litigant sample.
- The baseline female disadvantage is also larger in cases presided over by male judges.

Interpretation: The reform disciplines judges of both genders. The larger response among male judges is consistent with the idea that the pre-existing gender disparity was more pronounced under male judges.

Economic message: Monitoring does not only change one subgroup of judges; it shifts behavior across the judiciary, with stronger effects where the problem is larger.

5.10 Table 8: DID estimations with subsample of no litigant or lawyer appearing in court

Read:

- The authors restrict attention to cases where litigants or lawyers do not appear in court.
- Female $\times$ Intensity remains positive and significant in all four columns.
- Coefficients are 0.0193 and 0.0461 in the main sample, and 0.0337 and 0.0529 in the two-litigant sample.

Interpretation: These estimates weaken the alternative explanation that litigants or lawyers change their courtroom behavior when cameras are present. If the parties are absent, their live behavior cannot be the main channel.

Economic message: The reform's central effect appears to run through judges, not through litigants' or lawyers' strategic performance in the courtroom.

TABLE 8
DID estimations with subsample of no litigant or lawyer appearing in court

	Outcome Variable: Chances of Winning			
	All Litigants		Sub-sample: Two Litigants	
	Parties that do not appear:			
	Litigants (1)	Lawyers (2)	Litigants (3)	Lawyers (4)
Female $\times$ Intensity	0.0193 ^{***} (0.00316)	0.0461 ^{***} (0.00318)	0.0337 ^{***} (0.00442)	0.0529 ^{***} (0.00385)
Female	-0.00129 (0.000960)	-0.0205 ^{***} (0.00102)	-0.0128 ^{***} (0.00135)	-0.0231 ^{***} (0.00125)
Intensity	-0.0178 ^{***} (0.00502)	-0.0133 ^{***} (0.00623)	-0.0297 ^{***} (0.00681)	-0.00736 (0.0101)
Controls	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y
Court-Area FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y
Mean of outcome	0.748	0.771	0.737	0.738
N	1,935,756	2,044,952	782,414	1,251,355
R ²	0.415	0.554	0.550	0.651

Notes: This table presents DID estimation results with a subsample that none of litigant or lawyer appears in court. Both estimated coefficients remain significant with similar magnitude to those in the baseline case (see Table 2). Case-level control variables include the defendant gender, the number of judges, the number of plaintiffs and defendants, the number of lawyers for plaintiffs and defendants, whether plaintiffs and defendants appear in court or not, instance, legal cost (log), and case type. Prefecture-level control variables include internet penetration rate, GDP per capita (log) and population (log). Standard errors in parentheses clustered at court level; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Next, we examine whether the reform influenced plaintiffs' case filing behaviours. The introduction of the broadcasting mechanism may have discouraged litigants with weaker cases from filing claims. If female plaintiffs, on average, had a lower threshold for case strength when filing—meaning they tended to file weaker cases compared to male plaintiff—this could explain their lower winning chances before the reform. Consequently, the reform might mechanically

6 Interpretive synthesis

6.1 What the paper shows

- Female plaintiffs are disadvantaged in pre-reform civil litigation outcomes.
- Increased trial-broadcast intensity reduces this disadvantage.
- The result survives rich controls, judge fixed effects, court-area fixed effects, time fixed effects, alternative samples, event-study checks, and Bartik IV estimation.
- Text evidence points to judges becoming more attentive to female plaintiffs and more effortful in legal reasoning.
- The effect persists in samples where litigants or lawyers are absent, supporting a judge-behavior channel.

6.2 Understand the mechanism

The reform raises the visibility of judges' conduct. Visibility can operate through image concerns, career concerns, and increased expected costs of biased or careless conduct. Under stronger public monitoring, judges may pay more equal attention to litigants, cite law more carefully, and draft more individualized decisions. These behavioral changes can reduce gender disparities in outcomes.

6.3 What the paper does not show

- It does not prove that every pre-reform gender gap is taste-based discrimination.
- It does not show that broadcasting is always welfare-improving; privacy costs may be large.
- It does not fully identify long-run effects after judges become accustomed to monitoring.
- It does not prove that surveillance solves all judicial bias; hidden relationships or non-observable identities may remain unaffected.

7 Short summary

- The paper studies whether courtroom transparency reduces gender disparities in civil litigation.
- It uses roughly 6.4 million Chinese civil judgments from 2014-2018.
- The key policy is the 2016 open justice reform requiring live trial broadcasts.
- The main outcome is plaintiff chances of winning.
- The treatment is court-area by quarter broadcast intensity.
- Before the reform, female plaintiffs are less likely to win than male plaintiffs.
- Higher broadcast intensity reduces the female plaintiff disadvantage.
- Event-study results show effects grow after reform adoption.
- Bartik IV estimates confirm the DID result.
- Text similarity analysis suggests judges previously paid less attention to female plaintiffs' claims.
- After the reform, this attention gap narrows.
- Judges cite more legal articles and write less standardized judgments under greater broadcast intensity.
- Male and female judges both respond, with larger effects for male judges.
- Effects remain when litigants or lawyers are absent, pointing to judge behavior as the main channel.
- The paper concludes that information technology can increase accountability in public institutions, while recognizing privacy concerns.

8 Conclusion

8.1 Core conclusion

Chen, Chen, and Yang show that courtroom transparency can reduce gender disparities in civil litigation. The live-broadcast reform makes judges more visible to the public and to judicial supervisors, and this increased visibility is associated with fairer outcomes for female plaintiffs.

8.2 Economic intuition

Judges operate with discretion and limited direct oversight. When their proceedings can be observed and archived, the cost of appearing biased or careless rises. Judges respond by giving more attention to female plaintiffs' statements, citing more law, and producing less formulaic reasoning. These changes reduce the female plaintiff disadvantage.

8.3 Policy implication

Information technology can improve accountability in public institutions when it makes decision-makers' conduct observable. However, transparency is not free: broadcasting legal proceedings may impose privacy costs on litigants. A well-designed reform must balance accountability with privacy protection.

8.4 Final takeaway

The paper's central lesson is that digital transparency can discipline discretion. In this setting, technology does not merely publish information; it changes the behaviour of public officials and improves the fairness of outcomes.